

MINGLANG YIN, Ph.D.

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AI · Scientific ML · Modeling · Statistics · Multimodal · High-Performance Computing

SUMMARY

ML/AI researcher with 10+ years modeling sparse, noisy, high-dimensional real-world data. Depth in predictive and probabilistic modeling, foundation models, survival analysis, and computational sciences. Hands-on high-performance computing and distributed GPU training. 20+ peer-reviewed AI/ML publications from computational physics to clinical medicine, including Nature-family journals, AAAI, and leading applied mathematics venues.

EDUCATION

Brown University

Providence, RI

Ph.D. (Biomedical Engineering) and **M.Sc.** (Fluid and Thermal Sciences)

2016 – 2022

Division of Applied Mathematics · Advisor: Prof. George Em Karniadakis

Thesis: Hybrid Computational-Machine Learning Models with Uncertainty Quantification for Aortic Dissection

Northwestern Polytechnical University

Xi'an, China

B.S. (Aeronautical Engineering)

2012 – 2016

TECHNICAL SKILLS

Programming: Python, PyTorch, C++, Matlab.

Numerical and Statistical Modeling: numerical PDEs, numerical methods, computational fluid/solid dynamics, multiscale modeling; Bayesian inference, survival analysis, Markov Chain Monte Carlo (MCMC), gradient boosting (XGBoost/LightGBM).

High-Performance Computing: distributed multi-GPU training, OpenMPI.

RESEARCH EXPERIENCE

Johns Hopkins University

Baltimore, MD

Senior Postdoctoral Researcher & AI Group Lead

Aug 2022 – Present

Advisors: Profs. Natalia Trayanova (Biomedical Engineering), Mauro Maggioni (Applied Mathematics & Statistics)

Time-Series Foundation Model for Partial Differential Equations (PDEs)

- Proposed a novel AI framework for solving PDEs on geometries and achieved 10,000× speedup over numerical solvers; accomplished 98% accuracy on terabytes of clinical data.
- Developed a Time-Series PDE Foundation Model for enhancing model generalizability on out-of-distribution (OOD) data; Validated across thousands of human hearts/interventions (>350 TB) with accuracy >95%.
- Expedited training using distributed multi-GPU training and Flash Attention.
- Published in *Nature Computational Science* and conferences (AAAI 2024, *Society for Industrial and Applied Mathematics (SIAM)*, and *Heart Rhythm Scientific Sessions*) Latest manuscript under review.

Multimodal AI for Forecasting and Survival Analysis

- Developed a multimodal AI framework fusing images with tabular covariates to predict time-to-event outcomes and forecast risk, outperforming the established benchmark by 15–35%; validated on held-out external datasets to quantify robustness under distribution shift.
- Led the study design and ideation; built the full statistical stack — hypothesis testing, feature selection, survival analysis, and model calibration — and deployed the model externally at a partner institution (NYU Langone Health).
- Published in *Nature Cardiovascular Research*, *JACC: Clinical Electrophysiology*, and *Journal of the American Heart Association*; presented in *Heart Rhythm Scientific Sessions*.

Feature Extraction, Representation Learning on High-Dimensional Data

- Developed a dimensionality reduction method for high-dimensional geometric data, improved forecasting baselines with the low-dimensional representations, and identified drivers of outcome.
- Secured competitive external grant funding, led study design, data curation and preprocessing, statistical analysis, large-scale computation, and visualization.
- Published in *Computers in Biology and Medicine*; second manuscript under review.

Brown University
Ph.D. Researcher, Division of Applied Mathematics

Providence, RI
2018 – 2022

Bayesian Inference & Uncertainty Quantification (UQ)

- Developed a generative ML model coupled with Markov Chain Monte Carlo (MCMC) for parameter estimation and uncertainty quantification on experimental data; outperformed Gaussian process regression and nonlinear least-squares baselines with a 50% reduction in RMSE.
- Built physics-informed ML models to infer latent parameters from sparse, noisy observations for ill-posed problems.
- Published in *AAAI*, *Journal of the Mechanics and Physics of Solids*, *Computer Methods in Applied Mechanics and Engineering*, and *Acta Mechanica Sinica*; presented at *American Physical Society*, *SIAM Conferences*.

High-Performance Computing

- Implemented a parallel computing solver module for nonlinear material simulation in a production-level C++ codebase (LAMMPS).
- Designed an ML–numerical solver coupling scheme achieving $>14,000\times$ speedup over direct simulation.
- Derived, implemented, and validated a new solver module for a 3D parallel fluid dynamics code (Nektar).
- Co-designed a hybrid GPU–CPU pipeline for multiscale simulation, demonstrating linear scaling.
- Published in *Computational Mechanics* and *Computer Methods in Applied Mechanics and Engineering*.

Surrogate Modeling for Nonlinear Systems

- Built reduced-order models to approximate expensive high-fidelity fluid and solid simulations at a fraction of the computational cost.
- Developed data-driven surrogates for forecasting nonlinear dynamical systems from simulations and observational data.
- Published in *Journal of the Royal Society Interface* and *Computer Methods in Applied Mechanics and Engineering*; presented at SIAM conferences.

SELECTED PUBLICATIONS

- **M. Yin** et al. Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation, *Journal of the American Heart Association*, 2026.
- C. Lai, **M. Yin** et al. *Multimodal AI to forecast arrhythmic death in hypertrophic cardiomyopathy*. *Nature Cardiovascular Research*, 2025. [[JHU Hub](#), [Johns Hopkins Medicine](#), [New York Magazine](#)]
- **M. Yin** et al. *A scalable framework for learning geometry-dependent solution operators of PDEs*. *Nature Computational Science*, 2024. [[99th percentile](#); [Communication of the ACM](#), [MIT Tech Review](#), [NSF News](#), [JHU Hub](#)]
- **M. Yin**, et al. *A generative modeling framework for inferring families of biomechanical constitutive laws in data-sparse regimes*. *Journal of the Mechanics and Physics of Solids*, 2023.
- E. Zhang, **M. Yin**, et al. Physics-informed neural networks for nonhomogeneous material identification in elasticity imaging, *AAAI Conference*, 2020

SELECTED GRANTS & AWARDS

- **AHA Rapid Impact Research Award** (Collaborator; based on my NIH K99 proposal) 2026
- **Best PhD Thesis Award** (Int. J. Numer. Methods Biomed. Eng.) 2024
- **Kenneth M. Rosen Fellowship** 2023
- **Chinese Government Award for Outstanding PhD Students Abroad** (Top 1/600 globally) 2022